

AI61002 : Deep Learning Foundations and Applications

Spring 2025

Assignment 2: CNN for Facial Expression Recognition

(20 Marks)

Due Date: 17/03/2025

Instructions:

Complete the tasks listed below, ensuring you follow the specified instructions for each. Use PyTorch for implementation and provide step-by-step outputs in a Jupyter Notebook (.ipynb file). All relevant code snippets, visualizations, and explanations must be included within the notebook. Submit the required files as per the submission guidelines provided.

Task 1 - 2 Marks

Load the FER-2013 dataset from torchvision (train and test split). If torchvision does not contain the complete dataset, download it from Kaggle in .csv format: [FER-2013 Dataset](#). Ensure proper preprocessing and normalization of the dataset.

Task 2 - 1 Marks

Divide the train split into training and validation partitions. Ensure randomness in the split to avoid bias. Define corresponding dataloaders for each partition to facilitate batch-wise data loading.

Task 3 - 1 Marks

Visualize the dataset by displaying 3 randomly selected samples from each class. Ensure labels are correctly assigned.

Task 4 - 5 Marks

Design a custom Convolutional Neural Network (CNN) model with the following properties:

- Use 2 convolutional layers followed by Global Average Pooling (GAP) and a 1x1 convolutional layer for classification.
- Implement residual connections between successive layers, i.e., 2 residual connections, one for each layer.(2M)

- Ensure the architecture is fully convolutional with no fully connected (FC) layer. The FC layer should be replaced by GAP and 1x1 convolution. (1M)
- The number of channels for the 2 layers should be [32,64]. (1M)
- Use Relu as an intermediate activation function and an appropriate output activation function. (1M)

Task 5 - 3 Marks

Train the custom CNN model using a suitable loss function and the Adam optimizer with a learning rate of 1×10^{-3} (1M). Track the performance by:

- Plotting the training and validation losses against the number of epochs. (2M)
- Ensuring appropriate hyperparameter tuning for optimal performance.

Task 6 - 1 Marks

Evaluate the trained model on the test set and report the following metrics:

- Accuracy
- Precision
- Recall
- Confusion Matrix

Task 7 - 2 Marks

Generate Class Activation Maps (CAMs) using Grad-CAM for 1 randomly selected sample from each class. Visualize and analyze the activation regions.

Task 8 - 5 Marks

Implement a targeted Fast Gradient Sign Method (FGSM) adversarial attack to modify test images originally classified as “sad” so that they are misclassified as “happy” by the trained model. (4 M)

Report the success rate of the attack, which is defined as the fraction of samples that were initially correctly classified as "sad" but misclassified as "happy" after the adversarial perturbation. (1 M)

Submission Guidelines

- The content that you submit must be your individual work.
- Submit both .py and .ipynb file formats. Both submissions are mandatory for credit.

- Name the files as "<roll_number>_assignment_2.py" and "<roll_number>_assignment_2.ipynb". Submit them in a single zip file named "<roll_number>_assignment_2.zip". For example, if your roll number is 23AI91R01, the submission file should be named 23AI91R01_assignment_2.zip. (Note: Do not submit .rar files.)
- Ensure all code is well-commented, and explanations are included using text cells in the Jupyter Notebook.
- Submit your assignment through Moodle before the deadline. Late submissions or email submissions will not be accepted.

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